No.	Solution
1(a)	$F = kx$ $(0.140)(9.81) = k(0.108 - 0.080)$ $k = 49.1\text{ N m}^{-1}$
1(b)(i)	$k = \dfrac{mg}{L_2 - L_1}$ $\dfrac{\Delta k}{k} = \dfrac{\Delta m}{m} + \dfrac{\Delta(L_2 - L_1)}{(L_2 - L_1)}$ $\dfrac{\Delta k}{k} = 0.01 + \dfrac{0.002}{0.028} = 0.081$ $\%\dfrac{\Delta k}{k} = 8.1\%$
1(b)(ii)	$\dfrac{\Delta k}{k} = 0.081$ $\Delta k = (0.081)(49.1) = 4\text{ N m}^{-1}$ $k = (49 \pm 4)\text{ N m}^{-1}$
1(c)(i)	The upthrust is equal to the change in the tension in the spring. $U = (49.1)(0.5 \times 10^{-2})$ $= 0.25\text{ N}$
1(c)(ii)	$U = \rho_l V g$ $U = \rho_l \left(\dfrac{m}{\rho_b}\right) g$ $0.25 = \rho_l \left(\dfrac{0.140}{7750}\right)(9.81)$ $\rho_l = 1410\text{ kg m}^{-3}$
2(a)(i)	$2\pi r$